



Basic Layers of the GI Wall

Histology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 10 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.

EASY

From lumen to outside, which is the correct order of the four tunics of the GI wall?

- A. Serosa/adventitia → mucosa → submucosa → muscularis externa
- B. Muscularis externa → submucosa → mucosa → serosa/adventitia
- C. Mucosa → submucosa → muscularis externa → serosa/adventitia
- D. Submucosa → mucosa → muscularis externa → serosa/adventitia
- E. Mucosa → muscularis externa → submucosa → serosa/adventitia

2.

EASY

Which three layers together make up the mucosa?

- A. Epithelium, submucosa, muscularis externa
- B. Lamina propria, submucosa, serosa
- C. Epithelium, lamina propria, submucosa
- D. Muscularis mucosae, muscularis externa, serosa
- E. Epithelium, lamina propria, muscularis mucosae

3.

EASY

Auerbach's (myenteric) plexus lies:

- A. between the two muscle layers of the muscularis externa
- B. within the submucosa itself
- C. in the serosa beneath the mesothelium
- D. between the epithelium and the lamina propria
- E. within the lamina propria of the mucosa

4.

EASY

Which epithelium lines the esophagus?

- A. Simple columnar with a brush border
- B. Transitional epithelium
- C. Pseudostratified ciliated columnar
- D. Stratified squamous non-keratinized
- E. Simple cuboidal

5.

MEDIUM

CASE

A cross-section shows a star-shaped lumen lined by stratified squamous non-keratinized epithelium. Mucous glands lie in the dense connective tissue beneath the muscularis mucosae.

Which organ is this?

- A. Duodenum
- B. Ascending colon
- C. Esophagus
- D. Stomach body
- E. Anal canal

6.

MEDIUM

Meissner's (submucosal) plexus is responsible for regulating:

- A. contraction and peristalsis of the muscle coat
- B. local blood flow, absorption and secretion
- C. keratinization of the surface epithelium
- D. lubrication of the peritoneal cavity surface
- E. the position of the mesenteric attachment

7.

MEDIUM

CASE

A section of a thoracic tube shows a muscularis externa containing a mixture of skeletal and smooth muscle fibres.

Which part of which organ is this?

- A. Esophagus, middle third
- B. Stomach body
- C. Jejunum
- D. Esophagus, upper third
- E. Esophagus, lower third

8.

MEDIUM

A serosa (rather than an adventitia) tells you the segment is:

- A. fixed outside the peritoneal cavity
- B. found only in the thorax
- C. suspended inside the peritoneal cavity
- D. lined by stratified squamous epithelium
- E. without a muscularis externa

9.

MEDIUM

CASE

At surgery a segment of colon is found fixed to the posterior abdominal wall, with no glistening mesothelial covering on its posterior surface.

Which segment is most consistent with this finding?

- A. Ileum
- B. Transverse colon
- C. Stomach
- D. Jejunum
- E. Ascending colon

10.

MEDIUM

CASE

A thin, incomplete layer of smooth muscle lies between a loose connective tissue layer containing glands and a deeper layer of dense connective tissue containing large vessels.

Which layer is this?

- A. Muscularis externa
- B. Muscularis mucosae
- C. Lamina propria
- D. Serosa
- E. Adventitia

11.

MEDIUM

Which part of the esophagus is covered by serosa, and why?

- A. none of it, because the esophagus never has serosa
- B. the middle third, where skeletal and smooth muscle mix
- C. the whole length, because it is a muscular tube
- D. the upper third, because it contains skeletal muscle
- E. the last 2–3 cm, as only that part is inside the peritoneal cavity

12.

MEDIUM

Which statement about the peritoneum is correct?

- A. Peritoneum surrounds only the stomach wall
- B. Visceral peritoneum is a form of adventitia
- C. The peritoneal cavity is a solid connective tissue space
- D. Visceral covers most digestive organs; parietal lines the body wall
- E. Parietal covers the digestive organs; visceral lines the body wall

13.

MEDIUM

CASE

A researcher destroys the ganglia lying between the inner circular and outer longitudinal muscle layers of a bowel segment.

Which function is most directly lost?

- A. absorption of nutrients into the mucosa
- B. secretion of digestive fluids and enzymes
- C. lubrication of the serosal surface
- D. coordinated contraction and peristalsis
- E. local blood flow to the mucosa

14.

MEDIUM

Which sequence describes the epithelium from the esophagus to the anus?

- A. stratified squamous → simple columnar → stratified squamous
- B. simple columnar → stratified squamous → simple columnar
- C. transitional epithelium throughout the whole tube
- D. stratified squamous throughout the whole tube
- E. simple columnar throughout the whole tube

15.

HARD

CASE

Two unlabelled cross-sections come from the same esophagus. In section 1 the entire muscularis externa is skeletal muscle; in section 2 it is entirely smooth muscle.

Which statement is correct?

- A. Section 1 is from the upper third and section 2 from the lower third
- B. Section 2 cannot be esophagus because it has no skeletal muscle
- C. Section 1 cannot be esophagus because skeletal muscle is never found in the GI wall
- D. Both must come from the middle third
- E. Section 1 is from the lower third and section 2 from the upper third

16.

HARD

CASE

A biopsy shows stratified squamous non-keratinized epithelium on one side of a transition. Beyond it the epithelium is columnar and groups of mucous glands fill the lamina propria.

Which junction is this, and which glands lie in its lamina propria?

- A. Pyloroduodenal junction; esophageal cardiac glands
- B. Oropharyngeal junction; labial glands
- C. Esophagogastric junction; Brunner's glands
- D. Anorectal junction; crypts of Lieberkühn
- E. Esophagogastric junction; esophageal cardiac glands

17.

HARD

CASE

A tubular segment has (i) an adventitia rather than a serosa, (ii) mucous glands in its submucosa, and (iii) skeletal muscle fibres in its muscularis externa.

Which segment best fits all three findings?

- A. Esophagus, abdominal portion (2–3 cm)
- B. Esophagus, thoracic portion
- C. Duodenum, most of its length
- D. Stomach, body region
- E. Descending colon, retroperitoneal part

18.

HARD

CASE

Damage confined to the submucosa removes Meissner's plexus from a bowel segment while the muscle layers and Auerbach's plexus stay intact.

What is the expected result?

- A. no effect, as the two plexuses duplicate each other
- B. loss of every function, as the plexuses form one network
- C. impaired blood flow, absorption and secretion; peristalsis intact
- D. loss of peristalsis, with secretion unchanged
- E. loss of mesothelial lubrication only

19.

HARD

Why is the muscularis mucosae counted as part of the mucosa and not as part of the muscularis externa?

- A. It lies between the two layers of the muscularis externa
- B. It is a thin, incomplete layer at the deep edge of the mucosa, above the submucosa
- C. It contains Auerbach's plexus between its layers
- D. It is skeletal muscle, unlike the muscularis externa
- E. It lies outside the serosa, next to the mesentery

20.

HARD

CASE

A student lists what she sees from the lumen outward: epithelium; lamina propria; muscularis mucosae; dense connective tissue containing ganglia; inner circular muscle; ganglia; outer longitudinal muscle; mesothelium on connective tissue.

Which interpretation is best supported?

- A. Both sets = Meissner's, as ganglia are always submucosal
- B. Mesothelium shows the segment is retroperitoneal
- C. The dense CT is lamina propria and its ganglia are extrinsic nerves
- D. First ganglia = Meissner's, second = Auerbach's; mesothelium = serosa, so intraperitoneal
- E. First ganglia = Auerbach's, second = Meissner's; mesothelium = adventitia

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	C	Mucosa is innermost, then submucosa, muscularis externa, and serosa/adventitia outermost.
2	E	The mucosa = lining epithelium + lamina propria + muscularis mucosae.
3	A	The myenteric plexus sits between the two muscle layers and controls their contraction.
4	D	The esophageal mucosa is lined by stratified squamous non-keratinized epithelium (protection against abrasion).
5	C	A stellate lumen, stratified squamous non-keratinized epithelium and submucosal mucous glands identify the esophagus.
6	B	Meissner's plexus = local blood flow, absorption and secretion; contraction/peristalsis = Auerbach's.
7	A	Skeletal only = upper third; mixed skeletal + smooth = middle third; smooth only = lower third.
8	C	Serosa (mesothelium + CT) covers intraperitoneal organs; adventitia replaces it where the organ is fixed outside the cavity.
9	E	Ascending (and descending) colon is fixed outside the cavity and carries adventitia; transverse colon, stomach, jejunum and ileum carry serosa.
10	B	An incomplete smooth-muscle sheet between the mucosa and the submucosa is the muscularis mucosae.
11	E	Only the portion within the abdominal (peritoneal) cavity has serosa; the rest has an adventitia.
12	D	Visceral = covers organs; parietal = lines the body wall; the cavity between lubricates and lets organs slide.
13	D	The plexus between the two muscle layers (Auerbach's) controls contraction, movement and peristalsis.
14	A	The lecture's GI-tube summary: stratified squamous (esophagus) → simple columnar (stomach → rectum) → stratified squamous (anus).
15	A	Skeletal muscle only = upper third; smooth muscle only = lower third of the esophagus.
16	E	Squamous-to-columnar change below the diaphragm with mucous glands in the lamina propria = esophagogastric junction, esophageal cardiac glands.
17	B	Adventitia = outside the peritoneal cavity; submucosal mucous glands + skeletal muscle = thoracic esophagus (duodenum has adventitia but no skeletal muscle).
18	C	Meissner's plexus (submucosa) handles blood flow/absorption/secretion; the intact myenteric plexus keeps the muscle coat working.
19	B	Position defines it: between mucosa and submucosa, with the whole submucosa separating it from the muscularis externa.
20	D	Ganglia in the submucosa = Meissner's; between the muscle layers = Auerbach's; mesothelium on CT = serosa = intraperitoneal.

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Written from the lecture slides only – check any answer you doubt against your lecture notes.